

Intelligence On Files

Cloud Run Hackathon: Building serverless applications

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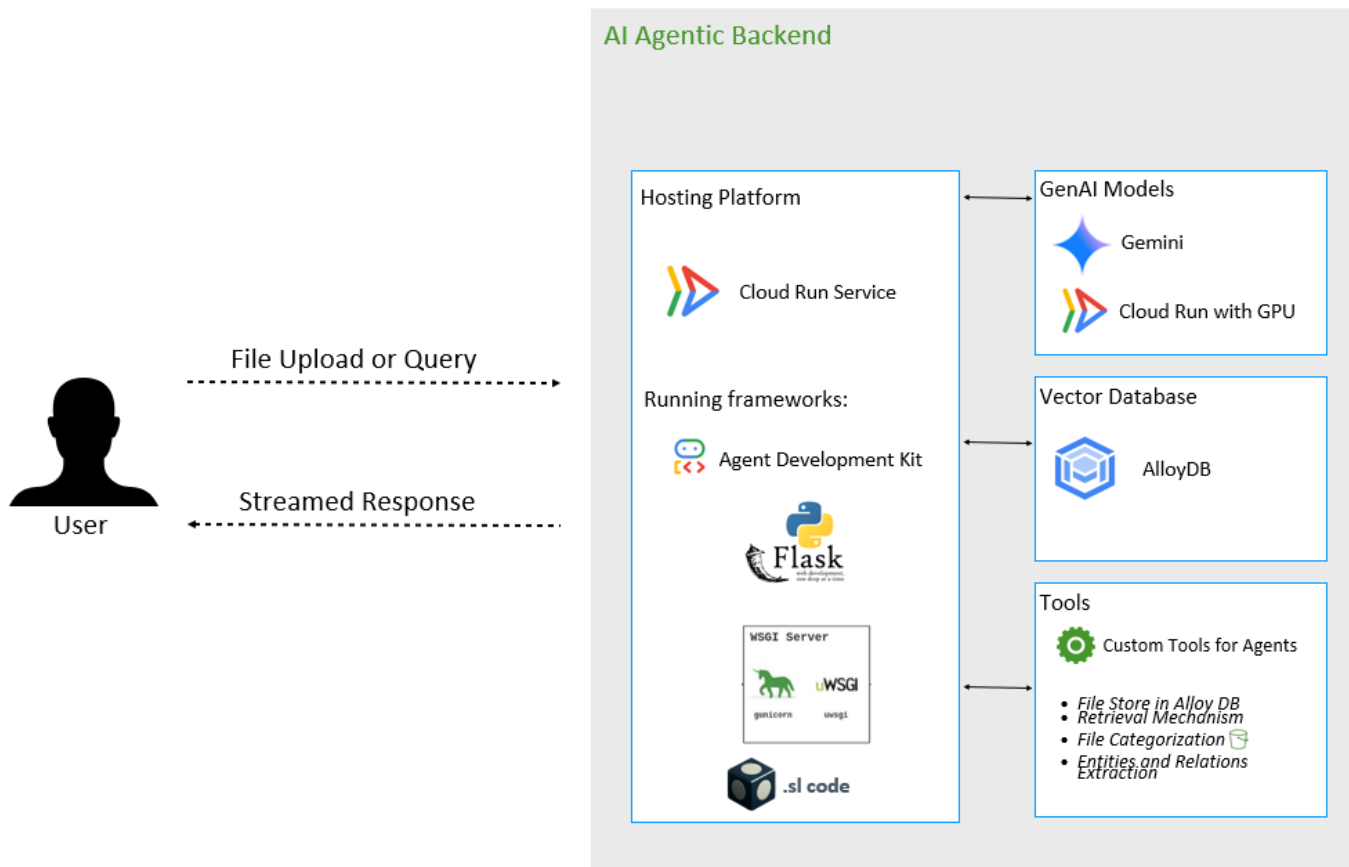


Figure 1: SeerVault's High Level Architecture

ABSTRACT

We present SeerVault, a production-grade, semantic-aware conversational cloud storage system that enables users to interact with their files using natural language, powered by a scalable multi-agent architecture on Google Cloud Run. Unlike traditional platforms such as Dropbox, Google Drive, or iCloud, which rely on manual navigation and filename based search, SeerVault allows users to *talk directly to their data*, issuing commands like "fetch my photos from 2014 in Paris with my family" or "I am looking for my tax-filing

documents from 2018". The system autonomously understands intent, retrieves files by semantic meaning, and executes multi-file operations server-side.

The Agents built with ADK performs the extraction and retrieval communication to serve the user with best results. The files and its semantic information is directly stored on AlloyDB helping in high-performance storage, indexing and retrieval.

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KEYWORDS

file storage, vector databases, ADK agents, domain specific languages, retrieval,

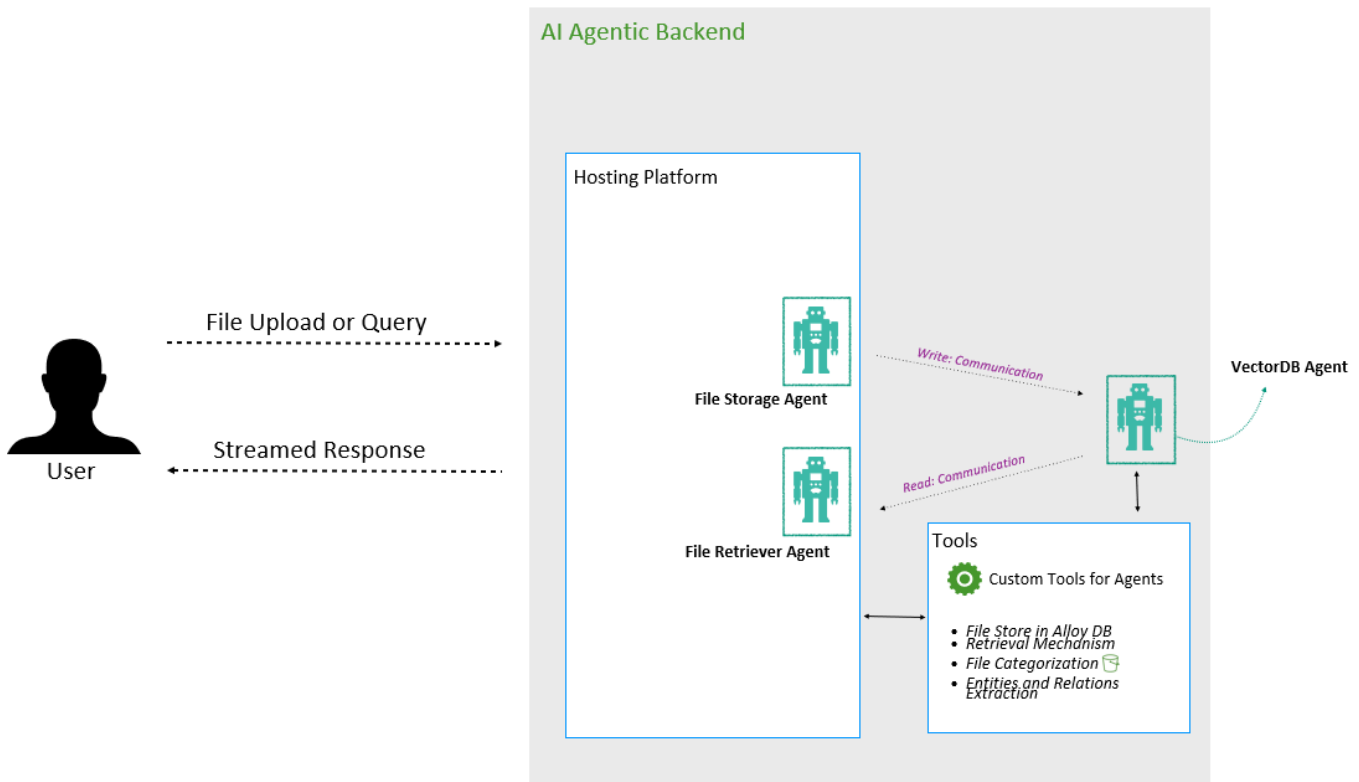


Figure 2: Agents in SeerVault

1 INTRODUCTION

Cloud storage has become a foundational infrastructure for personal and enterprise data management, yet contemporary platforms such as Dropbox, Google Drive, and iCloud largely function as passive repositories. These systems rely on hierarchical folder structures and filename-based search, requiring users to manually locate files, download content, and utilize external tools for analysis or processing. As data volumes increase and knowledge work spans diverse modalities and formats, this paradigm imposes significant cognitive and operational burdens, limiting efficiency and insight extraction.

In this work, we introduce SeerVault, a semantic-aware conversational cloud storage system that fundamentally redefines user interaction with digital content. SeerVault enables natural language queries over heterogeneous data, allowing users to issue commands such as “fetch my photos from 2014 in Paris with my family” or “consolidate all sales spreadsheets into a dashboard” without concern for file location or format. The system employs a dual-identity architecture: files are maintained as durable binary objects in Google Cloud Storage and as continuously updated semantic representations in Elastic Cloud with vector indices. Automated extraction pipelines process PDFs, spreadsheets, text files, and images to generate structured content, entity graphs, visual embeddings, and temporal scene descriptors. These semantic representations form a knowledge substrate upon which ADK-powered agents operate, performing multi-file retrieval, reasoning, and transformation tasks.

High-performance AlloyDB storage ensures scalable indexing and rapid query response.

SeerVault’s contribution lies in bridging the gap between natural language understanding, multi-modal semantic representation, and operational cloud storage. By embedding intelligence directly into the storage layer, the system eliminates manual download-edit-upload workflows, enabling users to interact with their data through meaningful intent rather than rigid file system abstractions. This work demonstrates a scalable architecture for semantic-aware cloud storage, highlighting the potential for AI-driven systems to transform digital content management.

2 ENGINEERING CHALLENGES

Building SeerVault required addressing seven interconnected challenges across the semantic search and multi-agent orchestration stack. Cross-modal semantic extraction demanded robust pipelines to parse PDFs, spreadsheets, text, and images while preserving context for embedding generation. Scalable embedding and vector indexing necessitated integration with Elastic Cloud vector indices to enable low-latency semantic retrieval over heterogeneous datasets. Multi-agent orchestration required designing a distributed ADK framework capable of concurrent agent coordination, fault tolerance, and intent-driven reasoning for multi-step query execution. High-performance metadata management in AlloyDB

balanced transactional integrity with asynchronously updated embeddings, maintaining consistency across relational and semantic layers. Serverless scaling on Cloud Run demanded automated resource strategies to optimize latency and cost under dynamic workloads for ingestion, retrieval, and reasoning. Unified cross-modal retrieval necessitated carefully designed embeddings and fusion strategies combining textual, visual, and structural semantics for accurate relevance ranking. Finally, asynchronous pipeline reliability required resilient background ingestion and embedding updates capable of handling partial failures, retries, and eventual consistency across distributed storage and indexing layers. These challenges were addressed through tight integration with Google Cloud services, careful system decomposition, and comprehensive error handling mechanisms.

SeerVault is implemented as a modular, full-stack web application that integrates user-facing interfaces with scalable cloud-native backend services. The system follows a standard Flask-based MVC structure, complemented by dedicated services for semantic extraction, multi-agent orchestration, and high-performance storage.

At the application layer, the `flask-app/` directory organizes the codebase as follows:

- `app/routes.py` defines the API endpoints for file upload, search, and archive operations, connecting the front-end interactions to backend services.
- `app/models.py` contains the database models and object abstractions, including metadata and semantic representations of files.
- `app/services/` implements modular utilities and connectors to the semantic processing pipelines, multi-agent orchestration engine (ADK), and cloud storage layers.
- `app/static/` holds the front-end assets: `css/style.css` for aesthetic styling and `js/main.js` for dynamic client-side logic including file uploads, search queries, and archive management.
- `app/templates/index.html` provides the main UI, enabling natural language interaction with stored files.

The Flask application is deployed using a WSGI entry point (`wsgi.py`) to allow scalable cloud deployment. Files are stored as durable binary objects in Google Cloud Storage, while semantic embeddings, structured metadata, and vector indices reside in Elastic Cloud. High-performance relational storage (AlloyDB) supports rapid indexing and retrieval.

A serverless, event-driven multi-agent layer orchestrates the interaction between user queries, semantic retrieval, reasoning, and visualization. This design ensures that SeerVault can process heterogeneous file formats at scale, enabling users to interact with their data through meaningful natural language commands without manual download, transformation, or application switching.

3 USE CASES (CONSUMER AND ENTERPRISE)

SeerVault supports a broad spectrum of real-world scenarios by enabling natural language interaction with cloud-stored files across consumer and enterprise contexts.

3.1 Consumer

For personal users, SeerVault eliminates the burden of manually navigating hierarchical folder structures and remembering filenames. Common tasks include:

- **Photo Discovery:** Queries such as “show me pictures from my 2014 trip to Paris with my family” retrieve content using visual embeddings and temporal metadata.
- **Document Retrieval:** Users can request “my tax documents from 2018” or “my apartment lease agreements” without remembering exact file locations.
- **Personal Organization:** Semantic clustering allows grouping receipts, medical records, or academic documents with high precision.
- **Cross-Modal Summaries:** The system can summarize collections of photos, extract text from PDFs, and generate descriptive insights.

3.2 Enterprise

In enterprise environments, SeerVault amplifies productivity and reduces operational friction across teams holding heterogeneous data:

- **Knowledge Management:** Employees can request information like “all design drafts referencing our Q3 release” or “contracts involving Vendor A in 2021.”
- **Compliance and Governance:** Auditors can ask for “all documents related to GDPR breaches” without manual crawling.
- **Multi-File Transformation:** SeerVault can orchestrate retrieval and aggregation tasks, such as “combine all regional sales spreadsheets into a single dashboard.”
- **Cross-Department Insight:** Natural language queries reveal hidden relationships across documents, image assets, and structured data sources.

By embedding semantic intelligence directly into the storage layer, SeerVault introduces a conversational interaction model that reduces cognitive friction while enabling richer insight extraction.

4 WHY .SL ?

Building multi-agent systems with Google’s Agent Development Kit (ADK) traditionally requires extensive boilerplate: instantiating models, registering tools, creating runners, managing sessions, and configuring message routing across hundreds of lines of Python code. Our Domain Specific Language (.sl) eliminates this complexity by allowing developers to express agent definitions declaratively focusing on *what* agents should do rather than *how* to configure them. Our DSL is purpose built for seamless ADK integration, where every construct maps directly to ADK primitives: agent definitions become Agent classes, tools become FunctionTool objects, and declarative flows become message routing infrastructure. Multi-agent coordination which is traditionally scattered across callbacks, threading, and manual queue management, becomes first-class syntax:

```
flow -> Analyst when "analysis_needed" as string
schedule every 30s send search to Researcher
```

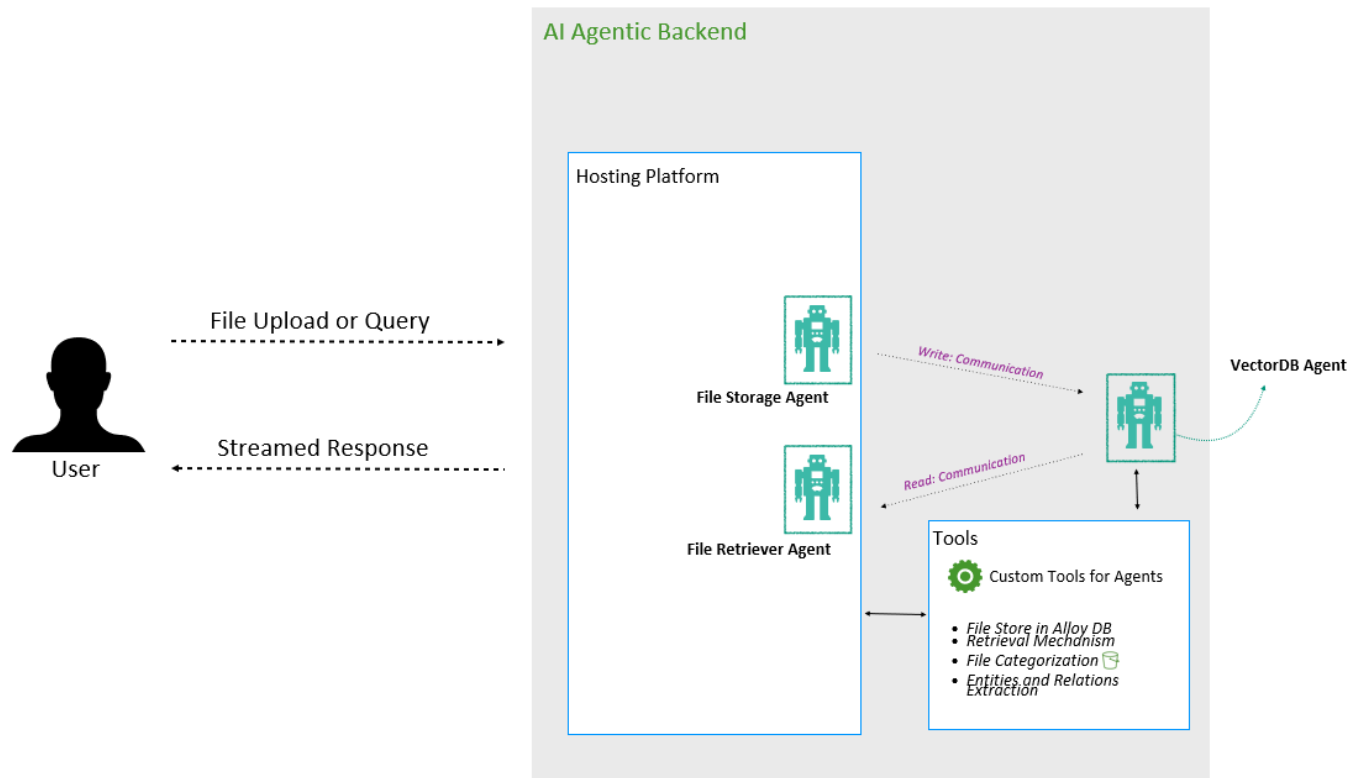


Figure 3: Agents in SeerVault

A single .sl file serves as the authoritative specification of system architecture, improving maintainability and enabling stakeholders to understand agent relationships, communication patterns, and scheduling at a glance. The generated code is production-ready and respects ADK conventions without manual refinement, while tool implementations can include raw Python for customization—avoiding framework lock-in. By bridging design diagrams (boxes and arrows) to executable code, the DSL ensures design intent survives implementation, reduces errors, and accelerates development cycles significantly.

5 CONCLUSION

SeerVault bridges the gap between traditional cloud storage and modern semantic understanding by allowing users to interact with their data through natural language. Leveraging multi-modal embeddings, distributed agent orchestration, and scalable serverless infrastructure, the system effectively transforms cloud storage from a passive repository into an intelligent, conversational interface. Early results demonstrate significant reductions in search time, greater utility of dormant files, and improved user experience for both consumer and enterprise use cases. As data continues to grow in volume and complexity, embedding semantic reasoning directly within storage systems represents a critical step toward more intuitive and powerful knowledge management.

6 FUTURE WORK

As SeerVault continues to evolve, several advanced research directions emerge that can further enhance its semantic-awareness, scalability, and integration fidelity. Future iterations will focus on more sophisticated cross-modal reasoning through hierarchical multi-vector representations and late-fusion attention mechanisms, enabling the system to disambiguate queries that combine visual context, temporal cues, and structured attributes. To reduce bandwidth and enhance data sovereignty, lightweight on-device embedding pipelines leveraging quantized transformer encoders may perform early-stage semantic extraction prior to cloud ingestion. At enterprise scale, federated semantic search across siloed data repositories and external knowledge graphs will require privacy-preserving techniques such as secure multi-party computation and homomorphic similarity search. Integrating temporal and session-based context models can enable dynamic re-ranking of retrieval results based on evolving user intent, while extending the agent framework toward autonomous workflow synthesis may support multi-step analytic tasks such as report generation, KPI aggregation, and cross-file summarization. Finally, future work must incorporate semantic-aware access control, applying attribute-based encryption and policy-level reasoning over embeddings to mitigate risk of latent information leakage. These directions collectively point toward a new class of scalable, privacy-preserving, and contextually adaptive storage systems capable of deeply understanding and operationalizing user intent in heterogeneous data environments.